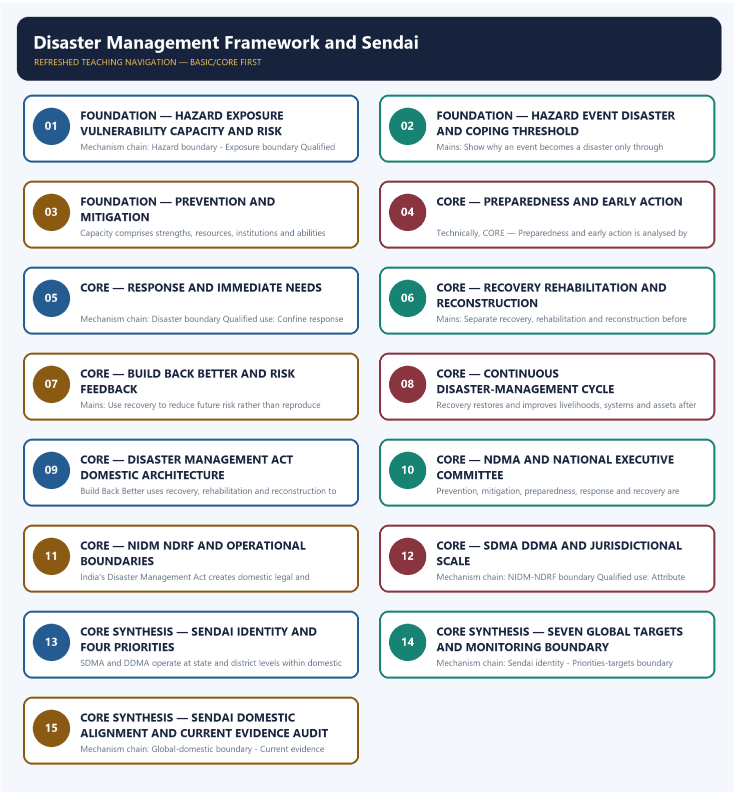
COMPLETE LEARNING SESSION

Disaster Management Framework and Sendai - Learner-v2 Refreshed

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing



Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

CONTENTS / SESSION INDEX

Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

Disaster Management Framework and Sendai - Learner-v2 Complete Learning Session

Authoring-only generation: 2026-09-03. No PDF was rendered and no tracker or index was mutated.

SOURCE, PROGRESSION AND CURRENT-LINKAGE AUDIT

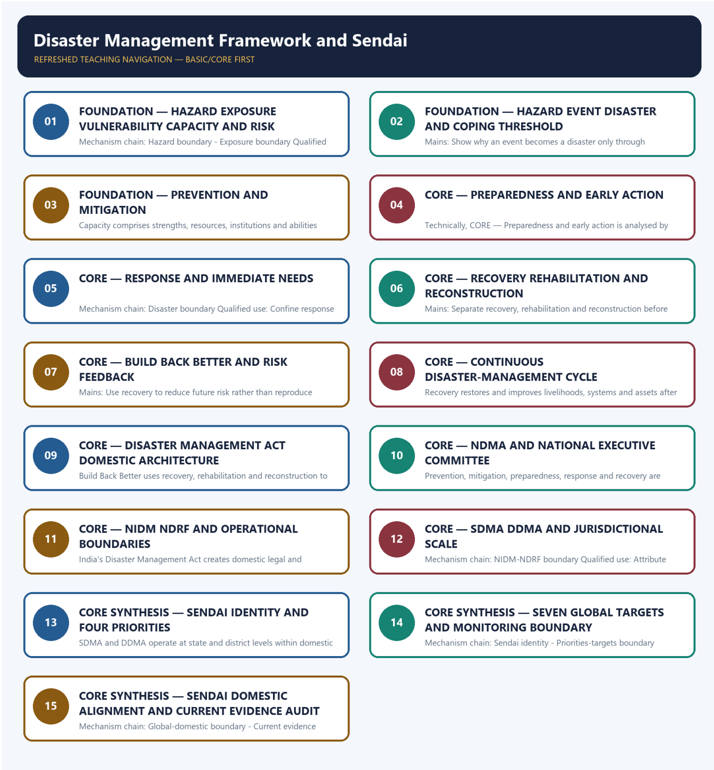
- **Generation date:** 2026-09-03.
- **Repository-first evidence:** the Basic owner is taught first and preserved in full; the Advanced owner is retained only in the optional final teaching block.
- **OCR evidence:** Repository Markdown was primary. OCR-searchable local official General Studies papers were used only to confirm printed routed demands. No answer key, marking scheme, unsupported page precision, ecological rate, species count, pollution standard, rule threshold, mission outcome or current status was inferred.
- **Qdrant:** not used; repository Markdown and available OCR context were sufficient.
- **PYQ integrity:** Audited ledgers route the verified 2024 resilience, Sendai-target and urban-flood demands plus related disaster-management questions. No objective key, loss figure, target progress or official model answer is inferred.
- **Live-link boundary:** UNDRR supplied substantive Sendai framework text. Indian official routes did not yield substantive current legal or plan text in this check, so no disaster loss, target progress, indicator, warning coverage, amended power or plan-status claim is asserted.
- **Fact/inference discipline:** no current-affairs item, PYQ wording, figure or quotation is invented.

LIVE OFFICIAL-SOURCE ATTEMPT LOG

The checks below were made on 2026-09-03. Substantive official text is used only for the proposition it supports. Stubs, access failures, unrelated pages and thin landing pages are recorded and are not converted into ecological or current-status claims.

- https://www.undrr.org/publication/sendai-framework-disaster-risk-reduction-2015-2030 - attempted 2026-09-03; substantive official text confirmed four priorities, seven targets, adoption context and the Build Back Better wording.
- https://nidm.gov.in/DMAct.asp - attempted 2026-09-03; the fetched route returned only contact information, so no statutory role, amendment or section claim was imported.
- https://ndma.gov.in/Governance/DM-Act-2005 - attempted 2026-09-03; the official route returned HTTP 404, so no current Act or institutional wording was imported.
- https://ndma.gov.in/ - attempted 2026-09-03; no dated plan revision, disaster loss, warning-coverage value or institutional outcome was imported from the general route.

BASIC LEARNING SESSION



Refreshed teaching navigation

Distinct embedded teaching-navigation image. The separate continuous Cārvāka-style flowchart package remains an independent at-a-glance artifact.

SESSION 1 - FOUNDATION - HAZARD EXPOSURE VULNERABILITY CAPACITY AND RISK

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk.

Technical definition: Mechanism chain: Hazard boundary - Exposure boundary Qualified use: Define risk through hazard, exposure, vulnerability and capacity.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Mechanism chain: Hazard boundary - Exposure boundary Qualified use: Define risk through hazard, exposure, vulnerability and capacity.

MUST-WRITE KEYWORDS

- **FOUNDATION**
- **Hazard exposure vulnerability capacity**
- **risk**
- **Mechanism chain**
- **Qualified use**
- **A hazard**

How to use them: Frame the answer through FOUNDATION; define Hazard exposure vulnerability capacity, connect risk with Mechanism chain to explain the mechanism, and use Qualified use for the decisive comparison or qualification.

VISUAL FIRST

```
HAZARD EXPOSURE VULNERABILITY CAPACITY AND RISK
01. Hazard boundary
    |
    v
02. Exposure boundary
BOUNDARY -> Do not merge hazard with disaster.
```

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility.

NAMED EVIDENCE AND MECHANISM

- A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk.
- Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility.

EXAMINER CAUTION

- Do not merge hazard with disaster.

EXAM LINK

- **Prelims:** Preserve the exact receiving medium, system boundary, measured parameter, actor, process, spatial and temporal scale, rule vintage and scientific or legal status; never turn a context-dependent pattern into a universal rule.
- **Mains:** Define risk through hazard, exposure, vulnerability and capacity.

MINI RECAP

- **Mechanism chain:** Hazard boundary -> Exposure boundary
- **Qualified use:** Define risk through hazard, exposure, vulnerability and capacity.

CLOSING RECALL FLOW - FOUNDATION - HAZARD EXPOSURE VULNERABILITY CAPACITY AND RISK

```
START / CONCEPT: FOUNDATION - Hazard exposure vulnerability capacity and risk
        |
        v
EXACT TERMS: FOUNDATION · Hazard exposure vulnerability capacity · risk · Mechanism chain ·
Qualified use · A hazard
        |
        v
MECHANISM / ARGUMENT: A hazard is a potentially damaging process, phenomenon or human activity; it
is not the same as a disaster or the complete measure of risk.
        |
        v
CONSEQUENCE / CONTRAST: Exposure identifies people, infrastructure, livelihoods, ecosystems and
assets located where hazards may occur; it does not describe their susceptibility.
        |
        v
UPSC TRAP / ANSWER-USE: Do not merge hazard with disaster.
        |
        v
ANSWER-GRABBING FORMULATION: Mechanism chain: Hazard boundary - Exposure boundary Qualified use:
Define risk through hazard, exposure, vulnerability and capacity.
```

SESSION 2 - FOUNDATION - HAZARD EVENT DISASTER AND COPING THRESHOLD

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Mechanism chain: Vulnerability boundary Qualified use: Show why an event becomes a disaster only through disruption and coping limits.

Technical definition: Mains: Show why an event becomes a disaster only through disruption and coping limits.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Mechanism chain: Vulnerability boundary Qualified use: Show why an event becomes a disaster only through disruption and coping limits.

MUST-WRITE KEYWORDS

- **FOUNDATION**
- **Hazard event disaster**
- **coping threshold**
- **Mechanism chain**
- **Qualified use**
- **Vulnerability**

How to use them: Frame the answer through FOUNDATION; define Hazard event disaster, connect coping threshold with Mechanism chain to explain the mechanism, and use Qualified use for the decisive comparison or qualification.

VISUAL FIRST

```
HAZARD EVENT DISASTER AND COPING THRESHOLD
01. Vulnerability boundary
BOUNDARY -> Do not merge exposure with vulnerability.
```

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure.

NAMED EVIDENCE AND MECHANISM

- Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure.

EXAMINER CAUTION

- Do not merge exposure with vulnerability.

EXAM LINK

- **Prelims:** Preserve the exact receiving medium, system boundary, measured parameter, actor, process, spatial and temporal scale, rule vintage and scientific or legal status; never turn a context-dependent pattern into a universal rule.
- **Mains:** Show why an event becomes a disaster only through disruption and coping limits.

MINI RECAP

- **Mechanism chain:** Vulnerability boundary
- **Qualified use:** Show why an event becomes a disaster only through disruption and coping limits.

CLOSING RECALL FLOW - FOUNDATION - HAZARD EVENT DISASTER AND COPING THRESHOLD

```
START / CONCEPT: FOUNDATION - Hazard event disaster and coping threshold
        |
        v
EXACT TERMS: FOUNDATION · Hazard event disaster · coping threshold · Mechanism chain · Qualified
use · Vulnerability
        |
        v
MECHANISM / ARGUMENT: Mains: Show why an event becomes a disaster only through disruption and
coping limits.
        |
        v
CONSEQUENCE / CONTRAST: Do not merge exposure with vulnerability.
        |
        v
UPSC TRAP / ANSWER-USE: Do not miss this limiting distinction: vulnerability is the set of physical,
 social, economic and environmental conditions that increase susceptibility...
        |
        v
ANSWER-GRABBING FORMULATION: Mechanism chain: Vulnerability boundary Qualified use: Show why an
event becomes a disaster only through disruption and coping limits.
```

SESSION 3 - FOUNDATION - PREVENTION AND MITIGATION

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Mechanism chain: Capacity boundary Qualified use: Separate avoiding risk from reducing unavoidable impact.

Technical definition: Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Do not omit capacity from a risk analysis.

MUST-WRITE KEYWORDS

- **FOUNDATION**
- **Prevention**
- **mitigation**
- **Mechanism chain**
- **Qualified use**
- **Capacity**

How to use them: Frame the answer through FOUNDATION; define Prevention, connect mitigation with Mechanism chain to explain the mechanism, and use Qualified use for the decisive comparison or qualification.

VISUAL FIRST

```
PREVENTION AND MITIGATION
01. Capacity boundary
BOUNDARY -> Do not omit capacity from a risk analysis.
```

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses.

NAMED EVIDENCE AND MECHANISM

- Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses.

EXAMINER CAUTION

- Do not omit capacity from a risk analysis.

EXAM LINK

- **Prelims:** Preserve the exact receiving medium, system boundary, measured parameter, actor, process, spatial and temporal scale, rule vintage and scientific or legal status; never turn a context-dependent pattern into a universal rule.
- **Mains:** Separate avoiding risk from reducing unavoidable impact.

MINI RECAP

- **Mechanism chain:** Capacity boundary
- **Qualified use:** Separate avoiding risk from reducing unavoidable impact.

CLOSING RECALL FLOW - FOUNDATION - PREVENTION AND MITIGATION

```
START / CONCEPT: FOUNDATION - Prevention and mitigation
        |
        v
EXACT TERMS: FOUNDATION · Prevention · mitigation · Mechanism chain · Qualified use · Capacity
        |
        v
MECHANISM / ARGUMENT: Mechanism chain: Capacity boundary Qualified use: Separate avoiding risk from
reducing unavoidable impact.
        |
        v
CONSEQUENCE / CONTRAST: Capacity comprises strengths, resources, institutions and abilities
available to manage and reduce risk; low capacity can magnify vulnerability and losses.
        |
        v
UPSC TRAP / ANSWER-USE: Do not miss this limiting distinction: separate avoiding risk from reducing
unavoidable impact.
        |
        v
ANSWER-GRABBING FORMULATION: Do not omit capacity from a risk analysis.
```

SESSION 4 - CORE - PREPAREDNESS AND EARLY ACTION

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Mechanism chain: Risk relation Qualified use: Place warnings, plans, drills and readiness before the event.

Technical definition: Technically, CORE - Preparedness and early action is analysed by relating Preparedness to early action, then testing the relationship through Mechanism chain and Qualified use.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Do not infer risk from hazard intensity alone.

MUST-WRITE KEYWORDS

- **Preparedness**
- **early action**
- **Mechanism chain**
- **Qualified use**
- **Risk**
- **Qualified**

How to use them: Frame the answer through Preparedness; define early action, connect Mechanism chain with Qualified use to explain the mechanism, and use Risk for the decisive comparison or qualification.

VISUAL FIRST

```
PREPAREDNESS AND EARLY ACTION
01. Risk relation
BOUNDARY -> Do not infer risk from hazard intensity alone.
```

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone.

NAMED EVIDENCE AND MECHANISM

- Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone.

EXAMINER CAUTION

- Do not infer risk from hazard intensity alone.

EXAM LINK

- **Prelims:** Preserve the exact receiving medium, system boundary, measured parameter, actor, process, spatial and temporal scale, rule vintage and scientific or legal status; never turn a context-dependent pattern into a universal rule.
- **Mains:** Place warnings, plans, drills and readiness before the event.

MINI RECAP

- **Mechanism chain:** Risk relation
- **Qualified use:** Place warnings, plans, drills and readiness before the event.

CLOSING RECALL FLOW - CORE - PREPAREDNESS AND EARLY ACTION

```
START / CONCEPT: CORE - Preparedness and early action
        |
        v
EXACT TERMS: Preparedness · early action · Mechanism chain · Qualified use · Risk · Qualified
        |
        v
MECHANISM / ARGUMENT: Mechanism chain: Risk relation Qualified use: Place warnings, plans, drills
and readiness before the event.
        |
        v
CONSEQUENCE / CONTRAST: The resulting consequence is that do not infer risk from hazard intensity
alone.
        |
        v
UPSC TRAP / ANSWER-USE: Do not miss this limiting distinction: do not infer risk from hazard
intensity alone.
        |
        v
ANSWER-GRABBING FORMULATION: Do not infer risk from hazard intensity alone.
```

SESSION 5 - CORE - RESPONSE AND IMMEDIATE NEEDS

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster.

Technical definition: Mechanism chain: Disaster boundary Qualified use: Confine response to immediate life-saving and basic-needs action.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster.

MUST-WRITE KEYWORDS

- **Response**
- **immediate needs**
- **Mechanism chain**
- **Qualified use**
- **A disaster**
- **Disaster**

How to use them: Frame the answer through Response; define immediate needs, connect Mechanism chain with Qualified use to explain the mechanism, and use A disaster for the decisive comparison or qualification.

VISUAL FIRST

```
RESPONSE AND IMMEDIATE NEEDS
01. Disaster boundary
BOUNDARY -> Do not call every hazard event a disaster.
```

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster.

NAMED EVIDENCE AND MECHANISM

- A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster.

EXAMINER CAUTION

- Do not call every hazard event a disaster.

EXAM LINK

- **Prelims:** Preserve the exact receiving medium, system boundary, measured parameter, actor, process, spatial and temporal scale, rule vintage and scientific or legal status; never turn a context-dependent pattern into a universal rule.
- **Mains:** Confine response to immediate life-saving and basic-needs action.

MINI RECAP

- **Mechanism chain:** Disaster boundary
- **Qualified use:** Confine response to immediate life-saving and basic-needs action.

CLOSING RECALL FLOW - CORE - RESPONSE AND IMMEDIATE NEEDS

```
START / CONCEPT: CORE - Response and immediate needs
        |
        v
EXACT TERMS: Response · immediate needs · Mechanism chain · Qualified use · A disaster · Disaster
        |
        v
MECHANISM / ARGUMENT: Mechanism chain: Disaster boundary Qualified use: Confine response to
immediate life-saving and basic-needs action.
        |
        v
CONSEQUENCE / CONTRAST: Do not call every hazard event a disaster.
        |
        v
UPSC TRAP / ANSWER-USE: Do not miss this limiting distinction: a disaster is serious disruption and
loss that exceeds or severely tests the affected...
        |
        v
ANSWER-GRABBING FORMULATION: A disaster is serious disruption and loss that exceeds or severely
tests the affected community's ability to cope; a hazard event need not become a disaster.
```

SESSION 6 - CORE - RECOVERY REHABILITATION AND RECONSTRUCTION

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Mechanism chain: Prevention-mitigation distinction - Preparedness boundary Qualified use: Separate recovery, rehabilitation and reconstruction before applying Build Back Better.

Technical definition: Mains: Separate recovery, rehabilitation and reconstruction before applying Build Back Better.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Mechanism chain: Prevention-mitigation distinction - Preparedness boundary Qualified use: Separate recovery, rehabilitation and reconstruction before applying Build Back Better.

MUST-WRITE KEYWORDS

- **Recovery rehabilitation**
- **reconstruction**
- **Mechanism chain**
- **Qualified use**
- **Preparedness**
- **Separate**

How to use them: Frame the answer through Recovery rehabilitation; define reconstruction, connect Mechanism chain with Qualified use to explain the mechanism, and use Preparedness for the decisive comparison or qualification.

VISUAL FIRST

```
RECOVERY REHABILITATION AND RECONSTRUCTION
01. Prevention-mitigation distinction
    |
    v
02. Preparedness boundary
BOUNDARY -> Do not merge prevention with mitigation.
```

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response.

NAMED EVIDENCE AND MECHANISM

- Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented.
- Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response.

EXAMINER CAUTION

- Do not merge prevention with mitigation.

EXAM LINK

- **Prelims:** Preserve the exact receiving medium, system boundary, measured parameter, actor, process, spatial and temporal scale, rule vintage and scientific or legal status; never turn a context-dependent pattern into a universal rule.
- **Mains:** Separate recovery, rehabilitation and reconstruction before applying Build Back Better.

MINI RECAP

- **Mechanism chain:** Prevention-mitigation distinction -> Preparedness boundary
- **Qualified use:** Separate recovery, rehabilitation and reconstruction before applying Build Back Better.

CLOSING RECALL FLOW - CORE - RECOVERY REHABILITATION AND RECONSTRUCTION

```
START / CONCEPT: CORE - Recovery rehabilitation and reconstruction
        |
        v
EXACT TERMS: Recovery rehabilitation · reconstruction · Mechanism chain · Qualified use ·
Preparedness · Separate
        |
        v
MECHANISM / ARGUMENT: Mains: Separate recovery, rehabilitation and reconstruction before applying
Build Back Better.
        |
        v
CONSEQUENCE / CONTRAST: Preparedness develops knowledge, plans, warnings, training, stocks and
coordination for effective action before an event; it is not post-event response.
        |
        v
UPSC TRAP / ANSWER-USE: Do not merge prevention with mitigation.
        |
        v
ANSWER-GRABBING FORMULATION: Mechanism chain: Prevention-mitigation distinction - Preparedness
boundary Qualified use: Separate recovery, rehabilitation and reconstruction before applying Build
Back Better.
```

SESSION 7 - CORE - BUILD BACK BETTER AND RISK FEEDBACK

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Mechanism chain: Response boundary Qualified use: Use recovery to reduce future risk rather than reproduce vulnerability.

Technical definition: Mains: Use recovery to reduce future risk rather than reproduce vulnerability.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Mechanism chain: Response boundary Qualified use: Use recovery to reduce future risk rather than reproduce vulnerability.

MUST-WRITE KEYWORDS

- **Build Back Better**
- **risk feedback**
- **Mechanism chain**
- **Qualified use**
- **Response**
- **Qualified**

How to use them: Frame the answer through Build Back Better; define risk feedback, connect Mechanism chain with Qualified use to explain the mechanism, and use Response for the decisive comparison or qualification.

VISUAL FIRST

```
BUILD BACK BETTER AND RISK FEEDBACK
01. Response boundary
BOUNDARY -> Do not call preparedness a response activity.
```

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster.

NAMED EVIDENCE AND MECHANISM

- Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster.

EXAMINER CAUTION

- Do not call preparedness a response activity.

EXAM LINK

- **Prelims:** Preserve the exact receiving medium, system boundary, measured parameter, actor, process, spatial and temporal scale, rule vintage and scientific or legal status; never turn a context-dependent pattern into a universal rule.
- **Mains:** Use recovery to reduce future risk rather than reproduce vulnerability.

MINI RECAP

- **Mechanism chain:** Response boundary
- **Qualified use:** Use recovery to reduce future risk rather than reproduce vulnerability.

CLOSING RECALL FLOW - CORE - BUILD BACK BETTER AND RISK FEEDBACK

```
START / CONCEPT: CORE - Build Back Better and risk feedback
        |
        v
EXACT TERMS: Build Back Better · risk feedback · Mechanism chain · Qualified use · Response ·
Qualified
        |
        v
MECHANISM / ARGUMENT: Mains: Use recovery to reduce future risk rather than reproduce vulnerability.
        |
        v
CONSEQUENCE / CONTRAST: Do not call preparedness a response activity.
        |
        v
UPSC TRAP / ANSWER-USE: Do not miss this limiting distinction: use recovery to reduce future risk
rather than reproduce vulnerability.
        |
        v
ANSWER-GRABBING FORMULATION: Mechanism chain: Response boundary Qualified use: Use recovery to
reduce future risk rather than reproduce vulnerability.
```

SESSION 8 - CORE - CONTINUOUS DISASTER-MANAGEMENT CYCLE

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Mechanism chain: Recovery boundary Qualified use: Link every phase through feedback rather than a one-way cycle.

Technical definition: Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Mechanism chain: Recovery boundary Qualified use: Link every phase through feedback rather than a one-way cycle.

MUST-WRITE KEYWORDS

- **Continuous disaster-management cycle**
- **Mechanism chain**
- **Qualified use**
- **Recovery**
- **Qualified**
- **CORE - Continuous disaster-management cycle**

How to use them: Frame the answer through Continuous disaster-management cycle; define Mechanism chain, connect Qualified use with Recovery to explain the mechanism, and use Qualified for the decisive comparison or qualification.

VISUAL FIRST

```
CONTINUOUS DISASTER-MANAGEMENT CYCLE
01. Recovery boundary
BOUNDARY -> Do not call emergency response long-term recovery.
```

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components.

NAMED EVIDENCE AND MECHANISM

- Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components.

EXAMINER CAUTION

- Do not call emergency response long-term recovery.

EXAM LINK

- **Prelims:** Preserve the exact receiving medium, system boundary, measured parameter, actor, process, spatial and temporal scale, rule vintage and scientific or legal status; never turn a context-dependent pattern into a universal rule.
- **Mains:** Link every phase through feedback rather than a one-way cycle.

MINI RECAP

- **Mechanism chain:** Recovery boundary
- **Qualified use:** Link every phase through feedback rather than a one-way cycle.

CLOSING RECALL FLOW - CORE - CONTINUOUS DISASTER-MANAGEMENT CYCLE

```
START / CONCEPT: CORE - Continuous disaster-management cycle
        |
        v
EXACT TERMS: Continuous disaster-management cycle · Mechanism chain · Qualified use · Recovery ·
Qualified · CORE - Continuous disaster-management cycle
        |
        v
MECHANISM / ARGUMENT: Recovery restores and improves livelihoods, systems and assets after
emergency response, while rehabilitation and reconstruction are related but distinct components.
        |
        v
CONSEQUENCE / CONTRAST: Do not call emergency response long-term recovery.
        |
        v
UPSC TRAP / ANSWER-USE: Do not miss this limiting distinction: recovery restores and improves
livelihoods, systems and assets after emergency response.
        |
        v
ANSWER-GRABBING FORMULATION: Mechanism chain: Recovery boundary Qualified use: Link every phase
through feedback rather than a one-way cycle.
```

SESSION 9 - CORE - DISASTER MANAGEMENT ACT DOMESTIC ARCHITECTURE

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Mechanism chain: Build-Back-Better boundary Qualified use: Anchor domestic roles in the Disaster Management Act.

Technical definition: Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Mechanism chain: Build-Back-Better boundary Qualified use: Anchor domestic roles in the Disaster Management Act.

MUST-WRITE KEYWORDS

- **Disaster Management Act domestic architecture**
- **Mechanism chain**
- **Qualified use**
- **Build Back Better**
- **Qualified**
- **Anchor**

How to use them: Frame the answer through Disaster Management Act domestic architecture; define Mechanism chain, connect Qualified use with Build Back Better to explain the mechanism, and use Qualified for the decisive comparison or qualification.

VISUAL FIRST

```
DISASTER MANAGEMENT ACT DOMESTIC ARCHITECTURE
01. Build-Back-Better boundary
BOUNDARY -> Do not reduce recovery to rebuilding identical assets.
```

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability.

NAMED EVIDENCE AND MECHANISM

- Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability.

EXAMINER CAUTION

- Do not reduce recovery to rebuilding identical assets.

EXAM LINK

- **Prelims:** Preserve the exact receiving medium, system boundary, measured parameter, actor, process, spatial and temporal scale, rule vintage and scientific or legal status; never turn a context-dependent pattern into a universal rule.
- **Mains:** Anchor domestic roles in the Disaster Management Act.

MINI RECAP

- **Mechanism chain:** Build-Back-Better boundary
- **Qualified use:** Anchor domestic roles in the Disaster Management Act.

CLOSING RECALL FLOW - CORE - DISASTER MANAGEMENT ACT DOMESTIC ARCHITECTURE

```
START / CONCEPT: CORE - Disaster Management Act domestic architecture
        |
        v
EXACT TERMS: Disaster Management Act domestic architecture · Mechanism chain · Qualified use ·
Build Back Better · Qualified · Anchor
        |
        v
MECHANISM / ARGUMENT: Build Back Better uses recovery, rehabilitation and reconstruction to reduce
future risk rather than merely recreate pre-disaster vulnerability.
        |
        v
CONSEQUENCE / CONTRAST: Do not reduce recovery to rebuilding identical assets.
        |
        v
UPSC TRAP / ANSWER-USE: Do not miss this limiting distinction: anchor domestic roles in the
Disaster Management Act.
        |
        v
ANSWER-GRABBING FORMULATION: Mechanism chain: Build-Back-Better boundary Qualified use: Anchor
domestic roles in the Disaster Management Act.
```

SESSION 10 - CORE - NDMA AND NATIONAL EXECUTIVE COMMITTEE

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Mechanism chain: Cycle-continuum boundary Qualified use: Differentiate policy authority from executive coordination.

Technical definition: Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Mechanism chain: Cycle-continuum boundary Qualified use: Differentiate policy authority from executive coordination.

MUST-WRITE KEYWORDS

- **NDMA**
- **National Executive Committee**
- **Mechanism chain**
- **Qualified use**
- **Prevention, mitigation, preparedness, response and recovery**
- **Prevention**

How to use them: Frame the answer through NDMA; define National Executive Committee, connect Mechanism chain with Qualified use to explain the mechanism, and use Prevention, mitigation, preparedness, response and recovery for the decisive comparison or qualification.

VISUAL FIRST

```
NDMA AND NATIONAL EXECUTIVE COMMITTEE
01. Cycle-continuum boundary
BOUNDARY -> Do not make the disaster cycle a one-way sequence.
```

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence.

NAMED EVIDENCE AND MECHANISM

- Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence.

EXAMINER CAUTION

- Do not make the disaster cycle a one-way sequence.

EXAM LINK

- **Prelims:** Preserve the exact receiving medium, system boundary, measured parameter, actor, process, spatial and temporal scale, rule vintage and scientific or legal status; never turn a context-dependent pattern into a universal rule.
- **Mains:** Differentiate policy authority from executive coordination.

MINI RECAP

- **Mechanism chain:** Cycle-continuum boundary
- **Qualified use:** Differentiate policy authority from executive coordination.

CLOSING RECALL FLOW - CORE - NDMA AND NATIONAL EXECUTIVE COMMITTEE

```
START / CONCEPT: CORE - NDMA and National Executive Committee
        |
        v
EXACT TERMS: NDMA · National Executive Committee · Mechanism chain · Qualified use · Prevention,
mitigation, preparedness, response and recovery · Prevention
        |
        v
MECHANISM / ARGUMENT: Prevention, mitigation, preparedness, response and recovery are linked phases
with feedback; disaster management is not a linear one-time sequence.
        |
        v
CONSEQUENCE / CONTRAST: Do not make the disaster cycle a one-way sequence.
        |
        v
UPSC TRAP / ANSWER-USE: Do not miss this limiting distinction: prevention, mitigation, preparedness,
 response and recovery are linked phases with feedback.
        |
        v
ANSWER-GRABBING FORMULATION: Mechanism chain: Cycle-continuum boundary Qualified use: Differentiate
policy authority from executive coordination.
```

SESSION 11 - CORE - NIDM NDRF AND OPERATIONAL BOUNDARIES

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Mechanism chain: Domestic-law boundary - NDMA-NEC boundary Qualified use: Differentiate training and research from specialised response operations.

Technical definition: India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Mechanism chain: Domestic-law boundary - NDMA-NEC boundary Qualified use: Differentiate training and research from specialised response operations.

MUST-WRITE KEYWORDS

- **NIDM NDRF**
- **operational boundaries**
- **Mechanism chain**
- **Qualified use**
- **India's Disaster Management Act**
- **Sendai Framework**

How to use them: Frame the answer through NIDM NDRF; define operational boundaries, connect Mechanism chain with Qualified use to explain the mechanism, and use India's Disaster Management Act for the decisive comparison or qualification.

VISUAL FIRST

```
NIDM NDRF AND OPERATIONAL BOUNDARIES
01. Domestic-law boundary
    |
    v
02. NDMA-NEC boundary
BOUNDARY -> Do not derive domestic legal powers from Sendai.
```

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework.

NAMED EVIDENCE AND MECHANISM

- India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework.
- NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework.

EXAMINER CAUTION

- Do not derive domestic legal powers from Sendai.

EXAM LINK

- **Prelims:** Preserve the exact receiving medium, system boundary, measured parameter, actor, process, spatial and temporal scale, rule vintage and scientific or legal status; never turn a context-dependent pattern into a universal rule.
- **Mains:** Differentiate training and research from specialised response operations.

MINI RECAP

- **Mechanism chain:** Domestic-law boundary -> NDMA-NEC boundary
- **Qualified use:** Differentiate training and research from specialised response operations.

CLOSING RECALL FLOW - CORE - NIDM NDRF AND OPERATIONAL BOUNDARIES

```
START / CONCEPT: CORE - NIDM NDRF and operational boundaries
        |
        v
EXACT TERMS: NIDM NDRF · operational boundaries · Mechanism chain · Qualified use · India's
Disaster Management Act · Sendai Framework
        |
        v
MECHANISM / ARGUMENT: India's Disaster Management Act creates domestic legal and institutional
duties; it must not be treated as the legal implementation text of the Sendai Framework.
        |
        v
CONSEQUENCE / CONTRAST: NDMA is the national authority for policy and guidelines, while the
National Executive Committee supports coordination, implementation and national planning under the
domestic framework.
        |
        v
UPSC TRAP / ANSWER-USE: Do not derive domestic legal powers from Sendai.
        |
        v
ANSWER-GRABBING FORMULATION: Mechanism chain: Domestic-law boundary - NDMA-NEC boundary Qualified
use: Differentiate training and research from specialised response operations.
```

SESSION 12 - CORE - SDMA DDMA AND JURISDICTIONAL SCALE

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other.

Technical definition: Mechanism chain: NIDM-NDRF boundary Qualified use: Attribute state and district roles only to their owned jurisdiction.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other.

MUST-WRITE KEYWORDS

- **SDMA DDMA**
- **jurisdictional scale**
- **Mechanism chain**
- **Qualified use**
- **NIDM**
- **NDRF**

How to use them: Frame the answer through SDMA DDMA; define jurisdictional scale, connect Mechanism chain with Qualified use to explain the mechanism, and use NIDM for the decisive comparison or qualification.

VISUAL FIRST

```
SDMA DDMA AND JURISDICTIONAL SCALE
01. NIDM-NDRF boundary
BOUNDARY -> Do not merge NDMA with NEC.
```

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other.

NAMED EVIDENCE AND MECHANISM

- NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other.

EXAMINER CAUTION

- Do not merge NDMA with NEC.

EXAM LINK

- **Prelims:** Preserve the exact receiving medium, system boundary, measured parameter, actor, process, spatial and temporal scale, rule vintage and scientific or legal status; never turn a context-dependent pattern into a universal rule.
- **Mains:** Attribute state and district roles only to their owned jurisdiction.

MINI RECAP

- **Mechanism chain:** NIDM-NDRF boundary
- **Qualified use:** Attribute state and district roles only to their owned jurisdiction.

CLOSING RECALL FLOW - CORE - SDMA DDMA AND JURISDICTIONAL SCALE

```
START / CONCEPT: CORE - SDMA DDMA and jurisdictional scale
        |
        v
EXACT TERMS: SDMA DDMA · jurisdictional scale · Mechanism chain · Qualified use · NIDM · NDRF
        |
        v
MECHANISM / ARGUMENT: Mechanism chain: NIDM-NDRF boundary Qualified use: Attribute state and
district roles only to their owned jurisdiction.
        |
        v
CONSEQUENCE / CONTRAST: Do not merge NDMA with NEC.
        |
        v
UPSC TRAP / ANSWER-USE: Do not miss this limiting distinction: nIDM focuses on training, research
and capacity development.
        |
        v
ANSWER-GRABBING FORMULATION: NIDM focuses on training, research and capacity development, while
NDRF is a specialised response force; neither should be substituted for the other.
```

SESSION 13 - CORE SYNTHESIS - SENDAI IDENTITY AND FOUR PRIORITIES

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Mechanism chain: State-district boundary Qualified use: State Sendai's global non-binding identity before its priorities.

Technical definition: SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Mechanism chain: State-district boundary Qualified use: State Sendai's global non-binding identity before its priorities.

MUST-WRITE KEYWORDS

- **CORE SYNTHESIS**
- **Sendai identity**
- **four priorities**
- **Mechanism chain**
- **Qualified use**
- **SDMA**

How to use them: Frame the answer through CORE SYNTHESIS; define Sendai identity, connect four priorities with Mechanism chain to explain the mechanism, and use Qualified use for the decisive comparison or qualification.

VISUAL FIRST

```
SENDAI IDENTITY AND FOUR PRIORITIES
01. State-district boundary
BOUNDARY -> Do not merge NIDM training with NDRF operations.
```

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text.

NAMED EVIDENCE AND MECHANISM

- SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text.

EXAMINER CAUTION

- Do not merge NIDM training with NDRF operations.

EXAM LINK

- **Prelims:** Preserve the exact receiving medium, system boundary, measured parameter, actor, process, spatial and temporal scale, rule vintage and scientific or legal status; never turn a context-dependent pattern into a universal rule.
- **Mains:** State Sendai's global non-binding identity before its priorities.

MINI RECAP

- **Mechanism chain:** State-district boundary
- **Qualified use:** State Sendai's global non-binding identity before its priorities.

CLOSING RECALL FLOW - CORE SYNTHESIS - SENDAI IDENTITY AND FOUR PRIORITIES

```
START / CONCEPT: CORE SYNTHESIS - Sendai identity and four priorities
        |
        v
EXACT TERMS: CORE SYNTHESIS · Sendai identity · four priorities · Mechanism chain · Qualified use ·
SDMA
        |
        v
MECHANISM / ARGUMENT: SDMA and DDMA operate at state and district levels within domestic law; their
plans and implementation roles must be attributed to the applicable statutory text.
        |
        v
CONSEQUENCE / CONTRAST: Do not merge NIDM training with NDRF operations.
        |
        v
UPSC TRAP / ANSWER-USE: Do not miss this limiting distinction: state Sendai's global non-binding
identity before its priorities.
        |
        v
ANSWER-GRABBING FORMULATION: Mechanism chain: State-district boundary Qualified use: State Sendai's
global non-binding identity before its priorities.
```

SESSION 14 - CORE SYNTHESIS - SEVEN GLOBAL TARGETS AND MONITORING BOUNDARY

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency.

Technical definition: Mechanism chain: Sendai identity - Priorities-targets boundary Qualified use: List global targets separately from action priorities.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions.

MUST-WRITE KEYWORDS

- **CORE SYNTHESIS**
- **Seven global targets**
- **monitoring boundary**
- **Mechanism chain**
- **Qualified use**
- **2015-2030**

How to use them: Frame the answer through CORE SYNTHESIS; define Seven global targets, connect monitoring boundary with Mechanism chain to explain the mechanism, and use Qualified use for the decisive comparison or qualification.

VISUAL FIRST

```
SEVEN GLOBAL TARGETS AND MONITORING BOUNDARY
01. Sendai identity
    |
    v
02. Priorities-targets boundary
BOUNDARY -> Do not move national roles automatically to a DDMA.
```

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency. Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions.

NAMED EVIDENCE AND MECHANISM

- The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency.
- Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions.

EXAMINER CAUTION

- Do not move national roles automatically to a DDMA.

EXAM LINK

- **Prelims:** Preserve the exact receiving medium, system boundary, measured parameter, actor, process, spatial and temporal scale, rule vintage and scientific or legal status; never turn a context-dependent pattern into a universal rule.
- **Mains:** List global targets separately from action priorities.

MINI RECAP

- **Mechanism chain:** Sendai identity -> Priorities-targets boundary
- **Qualified use:** List global targets separately from action priorities.

CLOSING RECALL FLOW - CORE SYNTHESIS - SEVEN GLOBAL TARGETS AND MONITORING BOUNDARY

```
START / CONCEPT: CORE SYNTHESIS - Seven global targets and monitoring boundary
        |
        v
EXACT TERMS: CORE SYNTHESIS · Seven global targets · monitoring boundary · Mechanism chain ·
Qualified use · 2015-2030
        |
        v
MECHANISM / ARGUMENT: Mechanism chain: Sendai identity - Priorities-targets boundary Qualified use:
List global targets separately from action priorities.
        |
        v
CONSEQUENCE / CONTRAST: The Sendai Framework 2015-2030 is a global, voluntary and non-binding
disaster-risk-reduction framework, not a domestic statute or operational response agency.
        |
        v
UPSC TRAP / ANSWER-USE: Do not move national roles automatically to a DDMA.
        |
        v
ANSWER-GRABBING FORMULATION: Sendai has four priorities for action and seven global targets; the
priorities describe fields of action, while the targets describe global reduction or increase
directions.
```

SESSION 15 - CORE SYNTHESIS - SENDAI DOMESTIC ALIGNMENT AND CURRENT EVIDENCE AUDIT

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: CORE SYNTHESIS - Sendai domestic alignment and current evidence audit comprises CORE SYNTHESIS, Sendai domestic alignment and current evidence audit as its core connected dimensions.

Technical definition: Mechanism chain: Global-domestic boundary - Current evidence boundary Qualified use: Close with dated domestic law, plan, indicator and UNDRR evidence.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction.

MUST-WRITE KEYWORDS

- **CORE SYNTHESIS**
- **Sendai domestic alignment**
- **current evidence audit**
- **Mechanism chain**
- **Qualified use**
- **Subject**

How to use them: Frame the answer through CORE SYNTHESIS; define Sendai domestic alignment, connect current evidence audit with Mechanism chain to explain the mechanism, and use Qualified use for the decisive comparison or qualification.

VISUAL FIRST

```
SENDAI DOMESTIC ALIGNMENT AND CURRENT EVIDENCE AUDIT
01. Global-domestic boundary
    |
    v
02. Current evidence boundary
BOUNDARY -> Do not call Sendai a binding treaty.
```

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence.

NAMED EVIDENCE AND MECHANISM

- Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction.
- Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence.

EXAMINER CAUTION

- Do not call Sendai a binding treaty.

EXAM LINK

- **Prelims:** Preserve the exact receiving medium, system boundary, measured parameter, actor, process, spatial and temporal scale, rule vintage and scientific or legal status; never turn a context-dependent pattern into a universal rule.
- **Mains:** Close with dated domestic law, plan, indicator and UNDRR evidence.

MINI RECAP

- **Mechanism chain:** Global-domestic boundary -> Current evidence boundary
- **Qualified use:** Close with dated domestic law, plan, indicator and UNDRR evidence.

COMPLETE BASIC OWNER EVIDENCE BANK

> **Subject:** Environment and Ecology | **Tier:** Must-Do (foundation) | **GS Paper:** GS-III (Disaster Management) + Prelims. **Core area:** Disaster-risk-reduction governance architecture. **Grounded in:** Disaster Management Act, 2005 (India Code); Sendai Framework for Disaster Risk Reduction 2015-2030 (UNDRR); NDMA guidelines; audited UPSC Environment/Disaster Management PYQs (2024-2026 Prelims and 2024-2025 Mains). FACT: = source-grounded | ANALYSIS: = analytical linkage | CURRENT: = dated current-affairs anchor. *Companion:* `advanced/26_Disaster-Management-Framework-and-Sendai.md`*.*

1. Visual foundation

```
DISASTER MANAGEMENT CYCLE (four linked phases, not a single event response)
   MITIGATION -> PREPAREDNESS -> RESPONSE -> RECOVERY -> (feeds back into MITIGATION)

INDIA'S THREE-TIER INSTITUTIONAL STRUCTURE (Disaster Management Act, 2005)
   NDMA (National) -> chaired by the Prime Minister
   SDMA (State)     -> chaired by the Chief Minister
   DDMA (District)  -> chaired by the District Magistrate/Collector

SENDAI FRAMEWORK (2015-2030) - FOUR PRIORITIES FOR ACTION
   1. Understanding disaster risk
   2. Strengthening disaster-risk governance to manage risk
   3. Investing in disaster-risk reduction for resilience
   4. Enhancing disaster preparedness for effective response, and to
      "Build Back Better" in recovery, rehabilitation and reconstruction
```

Core proposition: Effective disaster management is a continuous cycle (mitigation, preparedness, response, recovery), not a single reactive event-response - India's Disaster Management Act, 2005 established a three-tier institutional structure (NDMA-SDMA- DDMA) that operationalises this cycle, aligned globally with the Sendai Framework's shift from disaster *management* to disaster *risk reduction* as the guiding philosophy.

2. Essential definitions

Concept	Exam-ready meaning
FACT: **Disaster Management Act, 2005**	India's principal law establishing the institutional framework (NDMA, SDMA, DDMA) for disaster prevention, mitigation, preparedness, response and recovery.
FACT: **National Disaster Management Authority (NDMA)**	Apex national body, chaired by the Prime Minister, laying down disaster-management policies and guidelines.
FACT: **State Disaster Management Authority (SDMA)**	State-level body, chaired by the Chief Minister, responsible for state-level disaster-management planning.
FACT: **District Disaster Management Authority (DDMA)**	District-level body, chaired by the District Magistrate/Collector, responsible for district-level plan implementation.
FACT: **Sendai Framework for Disaster Risk Reduction 2015-2030**	A voluntary, non-binding UN framework (successor to the Hyogo Framework for Action), adopted in Sendai, Japan, in 2015, guiding global disaster-risk-reduction efforts.
FACT: **"Build Back Better"**	A Sendai Framework principle emphasising that post-disaster recovery/reconstruction should reduce future risk and improve resilience, not merely restore pre-disaster conditions.

3. Topic mechanism

1. The Disaster Management Act, 2005 (enacted following the 2004 Indian Ocean tsunami's demonstration of gaps in India's disaster-response coordination) created a formal three-tier institutional structure - NDMA at the national level, SDMA at the state level, and DDMA at the district level - each responsible for disaster-management planning and coordination at its respective level.

2. The Act mandates a shift from a purely reactive, relief-and-response orientation (historically dominant in Indian disaster governance) toward a comprehensive approach covering the full cycle: prevention, mitigation, capacity-building, preparedness, response, and rehabilitation/reconstruction.

3. The Sendai Framework (2015-2030), adopted at the UN's Third World Conference on Disaster Risk Reduction in Sendai, Japan, succeeded the earlier Hyogo Framework for Action and articulated four priorities for action: understanding disaster risk; strengthening disaster-risk governance; investing in disaster-risk reduction for resilience; and enhancing preparedness for effective response paired with "Build Back Better" recovery.

4. India, as a Sendai Framework signatory, aligns its National Disaster Management Plan with Sendai's priorities, emphasising risk understanding (hazard mapping, vulnerability assessment), governance strengthening (institutional coordination), and resilience investment (disaster-resilient infrastructure) as core strategic pillars.

5. The Sendai Framework's emphasis on "Build Back Better" reflects a conceptual shift from viewing disaster recovery as simple restoration to viewing it as an opportunity to reduce future vulnerability - for example, rebuilding damaged infrastructure to higher resilience standards rather than replicating pre-disaster vulnerabilities.

4. Institutions and policy tools

- FACT: **NDMA:** apex national disaster-management policy body, chaired by the Prime Minister.
- FACT: **National Institute of Disaster Management (NIDM):** provides training, capacity-building and research support for disaster-management professionals.
- FACT: **National Disaster Response Force (NDRF):** specialised, trained force for disaster response operations.

- FACT: **India Meteorological Department (IMD):** provides early-warning data (cyclone, extreme-weather forecasting) central to disaster preparedness.

5. Indian applications and examples

- ANALYSIS: India's cyclone early-warning and evacuation systems (particularly on the east coast, e.g., Odisha's documented improvements since major historical cyclone events) illustrate substantial preparedness-capacity improvement over time, contributing to significantly reduced casualty figures in more recent major cyclone events compared to historical precedents.
- ANALYSIS: Post-disaster reconstruction efforts (e.g., following major flood or cyclone events) have increasingly incorporated resilience-focused rebuilding principles, reflecting the Sendai Framework's "Build Back Better" influence on Indian disaster-recovery practice.
- ANALYSIS: India's National Disaster Management Plan explicitly references alignment with the Sendai Framework's four priorities for action.

6. Must-Know Facts for Prelims

- FACT: The Disaster Management Act, 2005 established a three-tier structure: NDMA (chaired by the Prime Minister), SDMA (chaired by the Chief Minister) and DDMA (chaired by the District Magistrate/Collector).
- FACT: The Sendai Framework for Disaster Risk Reduction 2015-2030 was adopted in 2015 at the UN's Third World Conference on Disaster Risk Reduction in Sendai, Japan, succeeding the Hyogo Framework for Action.
- FACT: The Sendai Framework articulates four priorities for action: understanding disaster risk, strengthening disaster-risk governance, investing in disaster-risk reduction for resilience, and enhancing preparedness for effective response and "Build Back Better."
- FACT: Sendai also sets **seven global targets (A-G)** - substantially **reduce** (A) disaster mortality, (B) the number of affected people, (C) direct economic loss relative to global GDP, and (D) damage to critical infrastructure and disruption of basic services; and substantially **increase** (E) the number of countries with national and local DRR strategies, (F) international cooperation to developing countries, and (G) availability of and access to multi-hazard early warning systems and disaster risk information. ANALYSIS: Four priorities **and** seven targets - a question asking for "the global targets of the Sendai Framework" is asking for the second set, not the first.
- FACT: The **World Conference on Disaster Risk Reduction** has been held three times, each hosted by **Japan**: **Yokohama (1994)**, **Kobe (2005)** and **Sendai (2015)**. Yokohama produced the *Yokohama Strategy for a Safer World* (10 principles, arising from the mid-term review of the International Decade for Natural Disaster Reduction); Kobe produced the **Hyogo Framework for Action 2005-2015** with **five priorities** and launched an International Early Warning Programme; Sendai produced the current framework - the **first major agreement of the post-2015 development agenda**.
- FACT: India's **National Disaster Management Plan, 2016** was the country's first such plan. It is aligned with the Sendai Framework, the SDGs and the Paris Agreement, and organises action under **five thematic areas**: understanding risk; inter-agency coordination; investing in DRR through **structural** measures; investing in DRR through **non-structural** measures; and capacity development. ANALYSIS: A documented critique is that, unlike Sendai or the SDGs, the NDMP sets **no goals, targets, timeframes or funding projections** of its own.
- FACT: The **Prime Minister's Ten-Point Agenda on Disaster Risk Reduction** was enunciated at the **Asian Ministerial Conference on Disaster Risk Reduction (AMCDRR), New Delhi, 2016**; each point maps to one or more Sendai priorities for action.
- FACT: The disaster-management cycle comprises mitigation, preparedness, response and recovery as continuously linked phases, not a single reactive event.
- FACT: The National Disaster Response Force (NDRF) is India's specialised disaster-response operational force.

7. UPSC traps

- WRONG: Disaster management refers only to post-disaster relief and response. -> It encompasses the full cycle: mitigation, preparedness, response and recovery.
- WRONG: The Sendai Framework is a legally binding international treaty. -> It is a voluntary, non-binding UN framework guiding national disaster-risk-reduction policy.
- WRONG: NDMA is chaired by the Union Home Minister. -> It is chaired by the Prime Minister.
- WRONG: The Sendai Framework was the first global disaster-risk-reduction framework. -> It succeeded the earlier Hyogo Framework for Action (2005-2015), which itself followed the Yokohama Strategy (1994).
- WRONG: The Sendai Framework's "global targets" and its "priorities for action" are the same list. -> There are **four priorities for action** and **seven global targets (A-G)** - a distinction UPSC has tested directly.
- WRONG: India's National Disaster Management Plan sets its own quantified targets like Sendai does. -> The NDMP is aligned with Sendai and the SDGs but does **not** set its own goals, targets, timeframes or funding projections - a documented critique.
- WRONG: "Build Back Better" means restoring exactly pre-disaster conditions as quickly as possible. -> It specifically means rebuilding to reduce future vulnerability and improve resilience, not merely replicating the pre-disaster state.

8. CURRENT: Current anchor

- CURRENT: **Disaster Management (Amendment) Act, 2025 (Act 10 of 2025)** came into force on **9 April 2025**. It updates the statutory architecture, including disaster databases and provisions concerning national crisis/high- level bodies and urban authorities.
- FACT: Use `../../Disaster-Management/basic/02_Indian-Legal-and-Institutional-Architecture.md` for detail rather than duplicating it here. NIDM/official text.
- CURRENT: India's National Disaster Management Plan continues to be updated in alignment with the Sendai Framework's 2015-2030 priorities; verify the latest NDMA plan revision and any updated early-warning-system capability (e.g., for cyclones, floods, or heatwaves) against the most recent NDMA/IMD publication before citing specific implementation details.

ANALYSIS: **Interpretation caution:** casualty-reduction figures for specific disaster events (e.g., comparing cyclone-related fatalities across different years/events) should be attributed to their specific source and event for precise, defensible citation.

9. PYQ application

- FACT: **2024 GS-III direct PYQ (250 words):** “What is disaster resilience? How is it determined? Describe various elements of a resilience framework. Also mention the global targets of the Sendai Framework for Disaster Risk Reduction (2015-2030).” ANALYSIS: Three distinct demands. For "elements of a resilience framework", use the standard DRR fields of action: a **policy framework** backed by legal and institutional mechanisms; **risk assessment** based on hazard and community resilience; **risk awareness** among stakeholders and decision-makers; **implementation** through environmental management and urban planning; **early-warning systems**; and **use of knowledge** through informed stakeholder participation and accessible communication. Then list the **seven global targets (A-G)** - the question asks for targets, not the four priorities.
- FACT: **2024 GS-III direct PYQ (250 words):** “Flooding in urban areas is an emerging climate-induced disaster. Discuss the causes of this disaster. Mention the features of two major floods in the last two decades in India. Describe the policies and frameworks aimed at dealing with such floods.” ANALYSIS: The question demands **two named Indian urban floods with their features** - a factual requirement most answers skip. The documented challenges to cite are: inadequate comprehensive urban-flood risk assessment before mitigation planning; failure to map city-specific risk factors into development planning; weak inter-agency coordination; poor information sharing; disintegrated investment decisions; and lack of stakeholder consultation.

- ANALYSIS: Recurring Prelims pattern: correctly identify the NDMA-SDMA-DDMA chairing structure and the Sendai Framework's four priorities for action.
- ANALYSIS: Mains linkage: the "Build Back Better" principle is used to argue for resilience-focused (not merely restorative) post-disaster reconstruction policy.

10. Mains angles

- ANALYSIS: Argue that effective disaster governance requires treating mitigation and preparedness (proactive, risk-reduction-oriented) as equally important to response and recovery (reactive), reflecting the full disaster-management-cycle logic.
- ANALYSIS: Use the "Build Back Better" principle to argue that post-disaster reconstruction should be an opportunity for resilience investment, not simple restoration.
- ANALYSIS: Conclude with a risk-governance thesis: disaster resilience depends on institutional coordination (NDMA-SDMA-DDMA) working effectively at every level, especially the often under-resourced district level where actual implementation occurs.

> **Answer thesis:** Treat disaster management as a continuous, four-phase cycle (not a single reactive response), and use the Sendai Framework's four priorities and "Build Back Better" principle as the analytical benchmark for evaluating India's disaster-risk-governance effectiveness at every institutional tier (NDMA-SDMA-DDMA).

11. Probable questions

- ANALYSIS: **Prelims:** Identify the correct chairing structure of NDMA, SDMA and DDMA, and the Sendai Framework's four priorities for action.
- ANALYSIS: **Mains (10 marks):** Explain the "Build Back Better" principle and its significance for India's post-disaster reconstruction policy.
- ANALYSIS: **Mains (15 marks):** Discuss the institutional strengths and implementation challenges of India's three-tier disaster-management structure under the Disaster Management Act, 2005.

12. Study links

- FACT: Advanced companion: `advanced/26_Disaster-Management-Framework-and-Sendai.md`.
- FACT: `17_Climate-Change-Science-Greenhouse-Effect.md` - climate change's role in intensifying certain disaster-risk categories (extreme weather events).
- FACT: `24_Coastal-and-Marine-Ecology-CRZ-Blue-Economy.md` - coastal-disaster-risk linkages relevant to cyclone preparedness.
- FACT: `27_Environmental-Institutions-MoEFCC-CPCB-NBA-WII.md` - NDMA's institutional relationship with other environmental governance bodies.

13. Core answer architecture (10/15/20-mark support)

13.1 Direct demand A - resilience framework and Sendai targets (2024 GS-III, 15 marks)

Thesis: disaster resilience is the capacity of people, systems and infrastructure to anticipate, absorb, adapt and recover without recreating risk; it is determined by hazard exposure, vulnerability, coping capacity, institutions, information and the quality of recovery.

Answer spine: definition/measurement logic -> resilience elements (risk assessment and land-use; legal/institutional coordination; resilient infrastructure and finance; ecosystem buffers; early warning and inclusive participation; preparedness/Build Back Better) -> seven Sendai global targets **A-D reduce mortality, affected people, economic loss and critical-infrastructure/basic-service disruption; E-G increase national/local strategies, international cooperation and multi-hazard early-warning/risk-information access** -> DDMA/ULB implementation verdict. Do not substitute the four priorities for the seven targets.

13.2 Direct demand B - urban floods (2024 GS-III, 15 marks)

Demand part	Core material
Causes	Extreme rainfall/climate signal interacting with sealed surfaces, encroached wetlands/floodplains, inadequate or blocked drainage, construction/land-use decisions, upstream releases and tidal/backflow where relevant.
Two cases	**Mumbai 2005:** exceptionally intense rainfall met drainage, low-lying/reclaimed-land and waste-blockage vulnerabilities. **Chennai 2015:** northeast-monsoon extreme rainfall combined with altered wetlands/waterways and urban drainage/land-use stress. Use each as a causal feature, not a decorative name.
Frameworks	NDMA **Guidelines on Management of Urban Flooding (2010)**; floodplain/zoning and drainage-catchment design; CWC-IMD warning/nowcasting; ULB/EOC response; Urban Flood Risk Management Programme; wetland restoration and risk-sensitive urban planning.

250-word spine: climate-risk framing -> physical and governance causes -> two compact cases -> structural/non-structural/institutional tools -> conclusion: restore storage and govern land use before relying on emergency pumping.

13.3 General 10/15/20 architecture

Use the Sendai shift from response to risk reduction, then map NDMA-SDMA-DDMA responsibilities, show a named ecosystem-based measure (mangrove/wetland/forest), identify the district/ULB capacity gap and end with a targeted, inclusive resilience verdict. Disaster-Management `basic/08` supplies deeper flood evidence; this Core contains a complete answer route.

CLOSING RECALL FLOW - CORE SYNTHESIS - SENDAI DOMESTIC ALIGNMENT AND CURRENT EVIDENCE AUDIT

```
START / CONCEPT: CORE SYNTHESIS - Sendai domestic alignment and current evidence audit
        |
        v
EXACT TERMS: CORE SYNTHESIS · Sendai domestic alignment · current evidence audit · Mechanism chain
· Qualified use · Subject
        |
        v
MECHANISM / ARGUMENT: Mechanism chain: Global-domestic boundary - Current evidence boundary
Qualified use: Close with dated domestic law, plan, indicator and UNDRR evidence.
        |
        v
CONSEQUENCE / CONTRAST: Do not call Sendai a binding treaty.
        |
        v
UPSC TRAP / ANSWER-USE: Core proposition: Effective disaster management is a continuous cycle (
mitigation, preparedness, response, recovery), not a single reactive event-response - India's
Disaster Management Act, 2005 established a three-tier institutional structure (NDMA-SDMA DDMA)
that operationalises this cycle, aligned globally with the Sendai Framework's shift from disaster
management to disaster risk reduction as the guiding philosophy.
        |
        v
ANSWER-GRABBING FORMULATION: Sendai's global targets guide international monitoring, while Indian
institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not
erase this distinction.
```

BASIC MCQS / REMEDIATION

Q1. Which statement correctly identifies Hazard boundary?

A. A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. B. Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. C. Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. D. Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility.

Answer: A. Explanation: A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q2. Which option preserves the ecological boundary of Hazard boundary?

A. Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. B. A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. C. Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. D. Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure.

Answer: B. Explanation: A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q3. Which statement uses Hazard boundary without changing its scale, parameter or status?

A. Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. B. Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. C. A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. D. A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster.

Answer: C. Explanation: A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q4. Which option avoids the standard UPSC close-option trap about Hazard boundary?

A. Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. B. Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. C. A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster. D. A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk.

Answer: D. Explanation: A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q5. Which statement correctly identifies Exposure boundary?

A. Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. B. Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. C. Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. D. Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses.

Answer: A. Explanation: Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q6. Which option preserves the ecological boundary of Exposure boundary?

A. Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. B. Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. C. A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster. D. Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses.

Answer: B. Explanation: Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q7. Which statement uses Exposure boundary without changing its scale, parameter or status?

A. A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster. B. Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. C. Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. D. Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented.

Answer: C. Explanation: Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q8. Which option avoids the standard UPSC close-option trap about Exposure boundary?

A. A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster. B. Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. C. Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. D. Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility.

Answer: D. Explanation: Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q9. Which statement correctly identifies Vulnerability boundary?

A. Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. B. A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster. C. Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. D. Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses.

Answer: A. Explanation: Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q10. Which option preserves the ecological boundary of Vulnerability boundary?

A. Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. B. Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. C. A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster. D. Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented.

Answer: B. Explanation: Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q11. Which statement uses Vulnerability boundary without changing its scale, parameter or status?

A. Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. B. A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster. C. Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. D. Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response.

Answer: C. Explanation: Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q12. Which option avoids the standard UPSC close-option trap about Vulnerability boundary?

A. Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster. B. Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. C. Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. D. Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure.

Answer: D. Explanation: Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q13. Which statement correctly identifies Capacity boundary?

A. Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. B. Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. C. A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster. D. Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented.

Answer: A. Explanation: Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q14. Which option preserves the ecological boundary of Capacity boundary?

A. Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. B. Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. C. A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster. D. Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented.

Answer: B. Explanation: Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q15. Which statement uses Capacity boundary without changing its scale, parameter or status?

A. Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster. B. Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. C. Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. D. Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented.

Answer: C. Explanation: Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q16. Which option avoids the standard UPSC close-option trap about Capacity boundary?

A. Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. B. Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components. C. Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster. D. Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses.

Answer: D. Explanation: Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q17. Which statement correctly identifies Risk relation?

A. Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. B. Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. C. Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. D. A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster.

Answer: A. Explanation: Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q18. Which option preserves the ecological boundary of Risk relation?

A. Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. B. Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. C. Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster. D. Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response.

Answer: B. Explanation: Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q19. Which statement uses Risk relation without changing its scale, parameter or status?

A. Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. B. Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components. C. Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. D. Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster.

Answer: C. Explanation: Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q20. Which option avoids the standard UPSC close-option trap about Risk relation?

A. Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. B. Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster. C. Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components. D. Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone.

Answer: D. Explanation: Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q21. Which statement correctly identifies Disaster boundary?

A. A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster. B. Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. C. Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster. D. Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response.

Answer: A. Explanation: A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q22. Which option preserves the ecological boundary of Disaster boundary?

A. Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components. B. A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster. C. Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. D. Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster.

Answer: B. Explanation: A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q23. Which statement uses Disaster boundary without changing its scale, parameter or status?

A. Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. B. Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster. C. A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster. D. Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components.

Answer: C. Explanation: A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q24. Which option avoids the standard UPSC close-option trap about Disaster boundary?

A. Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components. B. Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. C. Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. D. A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster.

Answer: D. Explanation: A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q25. Which statement correctly identifies Prevention-mitigation distinction?

A. Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. B. Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. C. Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components. D. Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster.

Answer: A. Explanation: Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q26. Which option preserves the ecological boundary of Prevention-mitigation distinction?

A. Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. B. Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. C. Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components. D. Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster.

Answer: B. Explanation: Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q27. Which statement uses Prevention-mitigation distinction without changing its scale, parameter or status?

A. Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. B. Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components. C. Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. D. Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence.

Answer: C. Explanation: Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q28. Which option avoids the standard UPSC close-option trap about Prevention-mitigation distinction?

A. India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. B. Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. C. Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. D. Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented.

Answer: D. Explanation: Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q29. Which statement correctly identifies Preparedness boundary?

A. Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. B. Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster. C. Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. D. Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components.

Answer: A. Explanation: Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q30. Which option preserves the ecological boundary of Preparedness boundary?

A. Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components. B. Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. C. Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. D. Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence.

Answer: B. Explanation: Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q31. Which statement uses Preparedness boundary without changing its scale, parameter or status?

A. Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. B. Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. C. Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. D. India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework.

Answer: C. Explanation: Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q32. Which option avoids the standard UPSC close-option trap about Preparedness boundary?

A. India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. B. NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. C. Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. D. Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response.

Answer: D. Explanation: Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q33. Which statement correctly identifies Response boundary?

A. Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster. B. Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. C. Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components. D. Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability.

Answer: A. Explanation: Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q34. Which option preserves the ecological boundary of Response boundary?

A. Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. B. Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster. C. Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. D. India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework.

Answer: B. Explanation: Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q35. Which statement uses Response boundary without changing its scale, parameter or status?

A. Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. B. India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. C. Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster. D. NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework.

Answer: C. Explanation: Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q36. Which option avoids the standard UPSC close-option trap about Response boundary?

A. NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. B. India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. C. NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other. D. Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster.

Answer: D. Explanation: Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q37. Which statement correctly identifies Recovery boundary?

A. Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components. B. Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. C. India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. D. Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability.

Answer: A. Explanation: Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q38. Which option preserves the ecological boundary of Recovery boundary?

A. NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. B. Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components. C. India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. D. Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence.

Answer: B. Explanation: Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q39. Which statement uses Recovery boundary without changing its scale, parameter or status?

A. NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other. B. NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. C. Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components. D. India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework.

Answer: C. Explanation: Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q40. Which option avoids the standard UPSC close-option trap about Recovery boundary?

A. SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. B. NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. C. NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other. D. Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components.

Answer: D. Explanation: Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q41. Which statement correctly identifies Build-Back-Better boundary?

A. Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. B. Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. C. India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. D. NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework.

Answer: A. Explanation: Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q42. Which option preserves the ecological boundary of Build-Back-Better boundary?

A. NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. B. Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. C. NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other. D. India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework.

Answer: B. Explanation: Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q43. Which statement uses Build-Back-Better boundary without changing its scale, parameter or status?

A. SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. B. NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other. C. Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. D. NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework.

Answer: C. Explanation: Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q44. Which option avoids the standard UPSC close-option trap about Build-Back-Better boundary?

A. NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other. B. The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency. C. SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. D. Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability.

Answer: D. Explanation: Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q45. Which statement correctly identifies Cycle-continuum boundary?

A. Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. B. NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. C. India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. D. NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other.

Answer: A. Explanation: Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q46. Which option preserves the ecological boundary of Cycle-continuum boundary?

A. SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. B. Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. C. NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. D. NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other.

Answer: B. Explanation: Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q47. Which statement uses Cycle-continuum boundary without changing its scale, parameter or status?

A. SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. B. NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other. C. Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. D. The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency.

Answer: C. Explanation: Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q48. Which option avoids the standard UPSC close-option trap about Cycle-continuum boundary?

A. SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. B. Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions. C. The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency. D. Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence.

Answer: D. Explanation: Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q49. Which statement correctly identifies Domestic-law boundary?

A. India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. B. NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other. C. NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. D. SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text.

Answer: A. Explanation: India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q50. Which option preserves the ecological boundary of Domestic-law boundary?

A. SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. B. India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. C. The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency. D. NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other.

Answer: B. Explanation: India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q51. Which statement uses Domestic-law boundary without changing its scale, parameter or status?

A. Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions. B. SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. C. India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. D. The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency.

Answer: C. Explanation: India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q52. Which option avoids the standard UPSC close-option trap about Domestic-law boundary?

A. Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions. B. Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. C. The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency. D. India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework.

Answer: D. Explanation: India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q53. Which statement correctly identifies NDMA-NEC boundary?

A. NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. B. The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency. C. NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other. D. SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text.

Answer: A. Explanation: NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q54. Which option preserves the ecological boundary of NDMA-NEC boundary?

A. Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions. B. NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. C. The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency. D. SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text.

Answer: B. Explanation: NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q55. Which statement uses NDMA-NEC boundary without changing its scale, parameter or status?

A. Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions. B. The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency. C. NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. D. Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction.

Answer: C. Explanation: NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q56. Which option avoids the standard UPSC close-option trap about NDMA-NEC boundary?

A. Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence. B. Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. C. Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions. D. NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework.

Answer: D. Explanation: NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q57. Which statement correctly identifies NIDM-NDRF boundary?

A. NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other. B. SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. C. The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency. D. Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions.

Answer: A. Explanation: NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q58. Which option preserves the ecological boundary of NIDM-NDRF boundary?

A. Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. B. NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other. C. The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency. D. Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions.

Answer: B. Explanation: NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q59. Which statement uses NIDM-NDRF boundary without changing its scale, parameter or status?

A. Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence. B. Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. C. NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other. D. Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions.

Answer: C. Explanation: NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q60. Which option avoids the standard UPSC close-option trap about NIDM-NDRF boundary?

A. A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. B. Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence. C. Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. D. NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other.

Answer: D. Explanation: NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q61. Which statement correctly identifies State-district boundary?

A. SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. B. Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. C. The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency. D. Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions.

Answer: A. Explanation: SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q62. Which option preserves the ecological boundary of State-district boundary?

A. Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions. B. SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. C. Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence. D. Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction.

Answer: B. Explanation: SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q63. Which statement uses State-district boundary without changing its scale, parameter or status?

A. A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. B. Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. C. SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. D. Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence.

Answer: C. Explanation: SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q64. Which option avoids the standard UPSC close-option trap about State-district boundary?

A. Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence. B. Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. C. A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. D. SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text.

Answer: D. Explanation: SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q65. Which statement correctly identifies Sendai identity?

A. The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency. B. Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. C. Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence. D. Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions.

Answer: A. Explanation: The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q66. Which option preserves the ecological boundary of Sendai identity?

A. Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence. B. The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency. C. A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. D. Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction.

Answer: B. Explanation: The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q67. Which statement uses Sendai identity without changing its scale, parameter or status?

A. Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. B. A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. C. The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency. D. Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence.

Answer: C. Explanation: The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q68. Which option avoids the standard UPSC close-option trap about Sendai identity?

A. A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. B. Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. C. Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. D. The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency.

Answer: D. Explanation: The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q69. Which statement correctly identifies Priorities-targets boundary?

A. Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions. B. A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. C. Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. D. Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence.

Answer: A. Explanation: Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q70. Which option preserves the ecological boundary of Priorities-targets boundary?

A. Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence. B. Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions. C. Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. D. A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk.

Answer: B. Explanation: Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q71. Which statement uses Priorities-targets boundary without changing its scale, parameter or status?

A. A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. B. Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. C. Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions. D. Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility.

Answer: C. Explanation: Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q72. Which option avoids the standard UPSC close-option trap about Priorities-targets boundary?

A. Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. B. Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. C. Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. D. Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions.

Answer: D. Explanation: Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q73. Which statement correctly identifies Global-domestic boundary?

A. Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. B. Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence. C. A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. D. Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility.

Answer: A. Explanation: Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q74. Which option preserves the ecological boundary of Global-domestic boundary?

A. A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. B. Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. C. Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. D. Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure.

Answer: B. Explanation: Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q75. Which statement uses Global-domestic boundary without changing its scale, parameter or status?

A. Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. B. Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. C. Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. D. Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility.

Answer: C. Explanation: Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q76. Which option avoids the standard UPSC close-option trap about Global-domestic boundary?

A. Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. B. Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. C. Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. D. Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction.

Answer: D. Explanation: Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q77. Which statement correctly identifies Current evidence boundary?

A. Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence. B. Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. C. A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. D. Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility.

Answer: A. Explanation: Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q78. Which option preserves the ecological boundary of Current evidence boundary?

A. Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. B. Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence. C. Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. D. Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses.

Answer: B. Explanation: Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q79. Which statement uses Current evidence boundary without changing its scale, parameter or status?

A. Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. B. Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. C. Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence. D. Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone.

Answer: C. Explanation: Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q80. Which option avoids the standard UPSC close-option trap about Current evidence boundary?

A. A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster. B. Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. C. Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. D. Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence.

Answer: D. Explanation: Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

PYQS AND ANSWER PRACTICE

AUDITED DISASTER RISK, RESILIENCE, SENDAI AND URBAN-FLOOD PYQ OWNERSHIP

Audited ledgers route the verified 2024 resilience, Sendai-target and urban-flood demands plus related disaster-management questions. No objective key, loss figure, target progress or official model answer is inferred.

OWNER PYQ LEDGER EXTRACTS

9. PYQ application

- FACT: **2024 GS-III direct PYQ (250 words):** “What is disaster resilience? How is it determined? Describe various elements of a resilience framework. Also mention the global targets of the Sendai Framework for Disaster Risk Reduction (2015-2030).” ANALYSIS: Three distinct demands. For "elements of a resilience framework", use the standard DRR fields of action: a **policy framework** backed by legal and institutional mechanisms; **risk assessment** based on hazard and community resilience; **risk awareness** among stakeholders and decision-makers; **implementation** through environmental management and urban planning; **early-warning systems**; and **use of knowledge** through informed stakeholder participation and accessible communication. Then list the **seven global targets (A-G)** - the question asks for targets, not the four priorities.
- FACT: **2024 GS-III direct PYQ (250 words):** “Flooding in urban areas is an emerging climate-induced disaster. Discuss the causes of this disaster. Mention the features of two major floods in the last two decades in India. Describe the policies and frameworks aimed at dealing with such floods.” ANALYSIS: The question demands **two named Indian urban floods with their features** - a factual requirement most answers skip. The documented challenges to cite are: inadequate comprehensive urban-flood risk assessment before mitigation planning; failure to map city-specific risk factors into development planning; weak inter-agency coordination; poor information sharing; disintegrated investment decisions; and lack of stakeholder consultation.
- ANALYSIS: Recurring Prelims pattern: correctly identify the NDMA-SDMA-DDMA chairing structure and the Sendai Framework's four priorities for action.
- ANALYSIS: Mains linkage: the "Build Back Better" principle is used to argue for resilience-focused (not merely restorative) post-disaster reconstruction policy.

10. PYQ-based analytical application

- ANALYSIS: Prelims questions on the Sendai Framework's priorities or India's three-tier structure should be answered by applying the precise priority-order and chairing-structure facts.
- ANALYSIS: Mains answers on "strengthening disaster management in India" should explicitly engage the DDMA-capacity-gap critique and the ecosystem-based-risk-reduction integration point to demonstrate analytical depth beyond describing the institutional structure.

PYQ DEMAND CARD 1 - 2024 GS-III

Demand: Define disaster resilience, explain its determination and elements, and mention Sendai global targets.

Status: Verified direct Mains demand; four priorities are not substituted for seven targets.

Model solution: Hazard boundary: A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. **Exposure boundary:** Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. **Vulnerability boundary:** Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. **Capacity boundary:** Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. **Risk relation:** Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. **Build-Back-Better boundary:** Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. **Priorities-targets boundary:** Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions. **Global-domestic boundary:** Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. The answer therefore separates definition, mechanism, evidence and limitation, and does not infer an official model answer or objective key.

PYQ DEMAND CARD 2 - 2024 GS-III

Demand: Discuss causes, cases and policy frameworks for urban flooding in India.

Status: Verified direct Mains demand; no disaster-loss value is inferred.

Model solution: Hazard boundary: A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. **Exposure boundary:** Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. **Vulnerability boundary:** Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. **Capacity boundary:** Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. **Risk relation:** Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. **Prevention-mitigation distinction:** Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. **Preparedness boundary:** Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. **Domestic-law boundary:** India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. **State-district boundary:** SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. **Current evidence boundary:** Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence. The answer therefore separates definition, mechanism, evidence and limitation, and does not infer an official model answer or objective key.

ORIGINAL MAINS 1 - 10 MARKS

Question: Distinguish hazard, exposure, vulnerability, capacity, risk and disaster. Answer in about 150 words.

Model thesis: Claim: Hazard boundary. **Named evidence/example:** A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Exposure boundary. **Named evidence/example:** Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Vulnerability boundary. **Named evidence/example:** Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Capacity boundary. **Named evidence/example:** Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Risk relation. **Named evidence/example:** Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Disaster boundary. **Named evidence/example:** A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Claim -> named evidence -> analysis -> qualification:

- A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk.
- Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility.
- Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure.
- Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses.
- Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone.
- A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster.

Qualified conclusion: Claim: Hazard boundary. **Named evidence/example:** A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Exposure boundary. **Named evidence/example:** Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Vulnerability boundary. **Named evidence/example:** Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Capacity boundary. **Named evidence/example:** Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Risk relation. **Named evidence/example:** Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Disaster boundary. **Named evidence/example:** A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

ORIGINAL MAINS 2 - 10 MARKS

Question: Explain prevention, mitigation and preparedness as distinct phases. Answer in about 150 words.

Model thesis: Claim: Prevention-mitigation distinction. **Named evidence/example:** Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Preparedness boundary. **Named evidence/example:** Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Cycle-continuum boundary. **Named evidence/example:** Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Claim -> named evidence -> analysis -> qualification:

- Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented.
- Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response.
- Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence.

Qualified conclusion: Claim: Prevention-mitigation distinction. **Named evidence/example:** Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Preparedness boundary. **Named evidence/example:** Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Cycle-continuum boundary. **Named evidence/example:** Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

ORIGINAL MAINS 3 - 15 MARKS

Question: Explain response, recovery and Build Back Better. Answer in about 250 words.

Model thesis: Claim: Response boundary. **Named evidence/example:** Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Recovery boundary. **Named evidence/example:** Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Build-Back-Better boundary. **Named evidence/example:** Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Cycle-continuum boundary. **Named evidence/example:** Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Claim -> named evidence -> analysis -> qualification:

- Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster.
- Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components.
- Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability.
- Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence.

Qualified conclusion: Claim: Response boundary. **Named evidence/example:** Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Recovery boundary. **Named evidence/example:** Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Build-Back-Better boundary. **Named evidence/example:** Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Cycle-continuum boundary. **Named evidence/example:** Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

ORIGINAL MAINS 4 - 15 MARKS

Question: Map NDMA, NEC, NIDM, NDRF, SDMA and DDMA by role. Answer in about 250 words.

Model thesis: Claim: Domestic-law boundary. **Named evidence/example:** India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** NDMA-NEC boundary. **Named evidence/example:** NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** NIDM-NDRF boundary. **Named evidence/example:** NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** State-district boundary. **Named evidence/example:** SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Claim -> named evidence -> analysis -> qualification:

- India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework.
- NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework.
- NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other.
- SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text.

Qualified conclusion: Claim: Domestic-law boundary. **Named evidence/example:** India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** NDMA-NEC boundary. **Named evidence/example:** NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** NIDM-NDRF boundary. **Named evidence/example:** NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** State-district boundary. **Named evidence/example:** SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

ORIGINAL MAINS 5 - 20 MARKS

Question: Compare Sendai's global framework with India's domestic legal architecture. Answer in about 300 words.

Model thesis: Claim: Domestic-law boundary. **Named evidence/example:** India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** NDMA-NEC boundary. **Named evidence/example:** NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** State-district boundary. **Named evidence/example:** SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Sendai identity. **Named evidence/example:** The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Priorities-targets boundary. **Named evidence/example:** Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Global-domestic boundary. **Named evidence/example:** Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Claim -> named evidence -> analysis -> qualification:

- India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework.
- NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework.
- SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text.
- The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency.
- Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions.
- Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction.

Qualified conclusion: Claim: Domestic-law boundary. **Named evidence/example:** India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** NDMA-NEC boundary. **Named evidence/example:** NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** State-district boundary. **Named evidence/example:** SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Sendai identity. **Named evidence/example:** The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Priorities-targets boundary. **Named evidence/example:** Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Global-domestic boundary. **Named evidence/example:** Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

ORIGINAL MAINS 6 - 20 MARKS

Question: Design a risk-reduction framework using the full disaster-management continuum. Answer in about 300 words.

Model thesis: Claim: Hazard boundary. **Named evidence/example:** A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Exposure boundary. **Named evidence/example:** Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Vulnerability boundary. **Named evidence/example:** Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Capacity boundary. **Named evidence/example:** Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses.

Analysis: This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Risk relation. **Named evidence/example:** Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Prevention-mitigation distinction. **Named evidence/example:** Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Preparedness boundary. **Named evidence/example:** Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Response boundary. **Named evidence/example:** Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Recovery boundary. **Named evidence/example:** Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Build-Back-Better boundary. **Named evidence/example:** Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Current evidence boundary. **Named evidence/example:** Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Claim -> named evidence -> analysis -> qualification:

- A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk.
- Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility.
- Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure.
- Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses.
- Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone.
- Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented.

- Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response.
- Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster.
- Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components.
- Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability.
- Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence.

Qualified conclusion: Claim: Hazard boundary. **Named evidence/example:** A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Exposure boundary. **Named evidence/example:** Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Vulnerability boundary. **Named evidence/example:** Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Capacity boundary. **Named evidence/example:** Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Risk relation. **Named evidence/example:** Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; 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OPTIONAL ADVANCED DEPTH - NOT REQUIRED FOR A CORE ANSWER

> **Subject:** Environment and Ecology | **Tier:** Advanced | **GS Paper:** GS-III (Disaster Management) + Prelims. **Core area:** Disaster-risk-reduction governance architecture. **Grounded in:** Disaster Management Act, 2005 full text; Sendai Framework for Disaster Risk Reduction 2015-2030 (UNDRR) monitoring indicators; National Disaster Management Plan revisions; audited UPSC Environment/Disaster Management PYQs (2024-2026 Prelims and 2024-2025 Mains). FACT: = source-grounded | ANALYSIS: = inference/analysis | CURRENT: = dated current-affairs anchor. *Companion:* `basic/26_Disaster-Management-Framework-and-Sendai.md`.

1. The paradigm shift Sendai represents: from "disaster management" to "disaster risk reduction"

ANALYSIS: The single most important advanced-level conceptual point: the global disaster-policy field underwent a genuine paradigm shift - from the Yokohama Strategy (1994) and Hyogo Framework (2005-2015) era's focus on *managing* disasters as they occur, toward the Sendai Framework's (2015-2030) explicit focus on *reducing existing disaster risk* and *preventing the creation of new risk* through development planning itself (e.g., land-use planning, building codes, ecosystem-based approaches) - treating disaster risk as a systemic development-planning issue rather than solely an emergency-response issue. This reframing is why Sendai's four priorities emphasise "understanding risk" and "risk governance" ahead of "preparedness for effective response" - response capability remains essential, but is positioned as the framework's fourth priority, not its starting point.

2. India's DDMA-level implementation gap: the recurring, examinable weak link

1. FACT: The Disaster Management Act, 2005's three-tier structure places DDMA (District

Disaster Management Authority) as the primary implementing body closest to actual disaster events, chaired by the District Magistrate/Collector.

2. ANALYSIS: **Documented implementation critique:** district-level disaster-management capacity

(dedicated staffing, technical expertise, updated hazard/vulnerability mapping, and resourcing for mitigation infrastructure) has been repeatedly flagged by various parliamentary and expert reviews as the weakest link in India's disaster-governance chain - District Magistrates typically hold disaster-management responsibilities alongside numerous other administrative duties, without always having a dedicated, full-time, technically specialised disaster-management team comparable to state or national-level bodies.

3. ANALYSIS: **Analytical synthesis:** this creates a structural mismatch - the institutional tier

with the most detailed local knowledge and the fastest initial-response role (DDMA) is often the least resourced tier in practice, while national/state bodies (NDMA/SDMA) with greater resources and policy authority are geographically and administratively more distant from the actual disaster event - a recurring theme when Mains questions ask about strengthening India's disaster-governance architecture.

3. Ecosystem-based disaster-risk reduction: a cross-topic integration point

- ANALYSIS: The Sendai Framework and India's evolving disaster policy increasingly recognise ecosystem-based approaches - healthy mangroves buffering cyclone storm surge (Topic 24), intact wetlands absorbing flood peaks (Topic 07), and stable forest cover reducing landslide/erosion risk (Topics 03, 11-12) - as disaster-risk-reduction infrastructure in their own right, not merely as separate "environmental" concerns.

- ANALYSIS: **Analytical significance:** this framing directly integrates disaster-management

policy with the ecosystem-services concept introduced in Topic 01 - degrading a mangrove forest for short-term coastal development is not just a biodiversity loss but a disaster-risk-increasing decision, since it removes a "regulating service" (storm-surge buffering) that has quantifiable value in disaster-risk terms. A sophisticated Mains answer on disaster management should explicitly cite ecosystem-based measures alongside conventional "grey" (engineered) infrastructure as part of a comprehensive risk-reduction strategy.

4. Early-warning systems: documented success and continuing gaps

- FACT: India's cyclone early-warning and evacuation capability (led by IMD forecasting combined

with NDMA/state-government coordinated evacuation protocols) has been credited with substantially reducing cyclone-related casualties in more recent major events compared to historical precedents from decades earlier - a genuine institutional success story worth citing with specific event comparisons where reliably documented.

- ANALYSIS: **Continuing gaps:** early-warning and preparedness capability is markedly stronger for

well-studied, more predictable hazards (cyclones, seasonal floods) than for less predictable or more localised hazards (flash floods, urban flooding, landslides, and heatwave-specific early-action protocols), which have received comparatively newer and less mature institutional attention - heatwave action plans, for instance, are a more recently developed instrument compared to the decades-long evolution of cyclone- preparedness systems.

5. Data and conceptual limitations

- ANALYSIS: Comparative casualty-reduction claims across different disaster events (e.g., citing

that a specific recent cyclone caused markedly fewer deaths than a historical comparator) should always cite the specific events and official casualty figures being compared, since disaster-mortality data can vary in reporting completeness and definition across different events and time periods.

- ANALYSIS: District-level disaster-management capacity assessments (staffing, resourcing,

technical capability) are not comprehensively, uniformly published in a single national dataset; specific capacity claims should be attributed to the relevant parliamentary committee report, NDMA assessment, or academic study cited.

- ANALYSIS: Quantifying the specific risk-reduction value of ecosystem-based measures (e.g., exact

storm-surge-height reduction attributable to a specific mangrove belt) requires site-specific scientific studies rather than a single universal figure.

6. Recurring UPSC analytical tensions

Tension	Balanced framing
National/state disaster-policy authority vs district-level implementation capacity	Genuine disaster resilience requires resourcing the DDMA tier commensurate with its critical, closest-to-ground implementation role, not just strengthening national/state policy frameworks.
Engineered ("grey") disaster infrastructure vs ecosystem-based ("green") risk reduction	Both are valuable and increasingly viewed as complementary; ecosystem-based measures (mangroves, wetlands, forests) should be integrated into disaster-risk-reduction planning, not treated as a separate environmental-policy silo.
Mature hazard-specific preparedness (cyclones) vs newer, less-developed hazard preparedness (heatwaves, urban flooding)	India's disaster-preparedness capability is uneven across hazard types; newer instruments (heatwave action plans) require continued institutional development comparable to the decades-long maturation of cyclone-preparedness systems.

7. Must-Know Facts for Advanced Prelims

- FACT: The Sendai Framework represents a paradigm shift from "disaster management" (managing

events as they occur) to "disaster risk reduction" (preventing/reducing risk through development planning itself), succeeding the Hyogo Framework for Action (2005-2015).

- FACT: District Disaster Management Authorities (DDMAs), chaired by the District Magistrate/ Collector, have been repeatedly flagged in reviews as the least-resourced tier in India's three-tier disaster-governance structure despite their critical, closest-to-ground implementation role.

- FACT: Ecosystem-based disaster-risk reduction (mangroves, wetlands, forests as risk-reduction infrastructure) is an increasingly recognised complement to engineered disaster infrastructure under both Sendai Framework principles and evolving Indian policy.

- FACT: India's cyclone early-warning/evacuation capability has matured substantially further than its preparedness capability for hazards like heatwaves or urban flooding, which are comparatively newer areas of institutional focus.

8. Advanced Prelims traps

- WRONG: The Sendai Framework primarily emphasises improving emergency response capability as its first priority. -> Its first priority is "understanding disaster risk," reflecting the paradigm shift toward risk reduction and prevention rather than response-first framing.

- WRONG: India's three-tier disaster-management structure is uniformly well-resourced across all levels. -> DDMA (district level) has been repeatedly flagged as comparatively under- resourced relative to its critical implementation role.

- WRONG: Ecosystem-based disaster-risk reduction (mangroves, wetlands) is considered separate from and unrelated to formal disaster-management policy. -> It is increasingly integrated as a recognised risk-reduction approach under Sendai Framework principles and evolving Indian policy.

- WRONG: India's disaster-preparedness capability is uniformly mature across all hazard types. -> It is notably stronger for well-studied hazards like cyclones than for newer areas like heatwave action planning or urban-flood preparedness.

9. CURRENT: Current anchor - analytical use

CURRENT: **Disaster Management (Amendment) Act, 2025**, in force 9 April 2025, is the current legal update. Route institutional detail to `../../Disaster-Management/advanced/02_Indian-Legal-and-Institutional-Architecture.md` instead of duplicating it. NIDM/official text.

Verified current anchor	Topic-specific analytical use
CURRENT: India's National Disaster Management Plan, aligned with Sendai Framework priorities, continues to be updated (verify the latest NDMA plan revision and hazard-specific action-plan development, e.g., heatwave action plans, before citing specific current details).	Use as the current institutional baseline while pairing it with the DDMA-capacity-gap critique and the ecosystem-based-risk-reduction integration point for full analytical depth.
FACT: **The Hyogo-to-Sendai shift is the analytical core.** Hyogo (2005-2015) set **five priorities** framed around *managing disasters*; Sendai (2015-2030) set **four priorities plus seven measurable global targets** framed around *managing risk*, and extended explicitly to **"Build Back Better" in recovery, rehabilitation and reconstruction**.	The examinable proposition: Sendai's decisive innovation was not new priorities but **measurable targets with a baseline period**, converting DRR from exhortation into something auditable. India's NDMP inherits the priorities but not the targets - which is precisely where the accountability gap sits.
FACT: **Heatwave governance** is the fastest-moving gap: heatwaves are currently treated largely as **local** disasters, while their intensifying severity argues for a broader national approach; documented weaknesses include the absence of sub-district-level data to build separate urban and rural heat indices, thin intra-city ward-level vulnerability analysis, and the lack of a settled financing arrangement for Heat Action Plans.	The single best contemporary illustration of the **hazard-classification-drives-financing** problem: how a hazard is legally and administratively categorised determines which fund pays for it. Use alongside Topic 17's SLCF/urban-heat science for a science-to-governance chain.

Verified current anchor	Topic-specific analytical use
FACT: **Ecosystem-based disaster risk reduction is now explicit in Indian fiscal policy** - MISHTI (mangroves as surge buffers), NPCA-expanded wetlands as flood-moderation buffers, and the National Coastal Mission (Economic Survey 2025-26, Ch. 10).	Lets a disaster-management answer draw directly on Topics 01, 07 and 24: the cheapest disaster infrastructure is often an intact ecosystem, and the Government now says so in its own budget documents.

10. PYQ-based analytical application

- ANALYSIS: Prelims questions on the Sendai Framework's priorities or India's three-tier structure should be answered by applying the precise priority-order and chairing-structure facts.
- ANALYSIS: Mains answers on "strengthening disaster management in India" should explicitly engage the DDMA-capacity-gap critique and the ecosystem-based-risk-reduction integration point to demonstrate analytical depth beyond describing the institutional structure.

11. Mains-ready framework

Central thesis: India's disaster-governance architecture reflects the Sendai Framework's paradigm shift toward risk reduction and prevention, but its real effectiveness is constrained by a persistent capacity gap at the district level - the tier closest to actual disaster events - and by uneven institutional maturity across hazard types; genuine resilience requires resourcing DDMAs commensurately, integrating ecosystem-based risk- reduction measures alongside engineered infrastructure, and extending the same institutional maturity seen in cyclone preparedness to newer hazard categories like heatwaves and urban flooding.

1. Explain the Sendai Framework's paradigm shift from disaster management to disaster risk reduction, and its four priorities.
2. Present India's three-tier structure (NDMA-SDMA-DDMA) and the documented DDMA-capacity gap as the central implementation critique.
3. Integrate ecosystem-based risk-reduction measures (mangroves, wetlands, forests) as a cross-topic analytical point connecting disaster policy to ecosystem-services concepts.
4. Compare mature (cyclone) versus newer (heatwave, urban flooding) hazard-preparedness institutional development.
5. Conclude with a DDMA-resourcing and ecosystem-integration recommendation as the genuine determinants of improved disaster resilience.

12. Probable questions

- ANALYSIS: **Prelims:** Identify the Sendai Framework's four priorities for action in correct order and India's three-tier disaster-management chairing structure.
- ANALYSIS: **Mains (10 marks):** Why has the District Disaster Management Authority been identified as the weakest link in India's disaster-governance structure?
- ANALYSIS: **Mains (15 marks):** Discuss the role of ecosystem-based approaches (mangroves, wetlands, forests) in disaster-risk reduction, integrating this with India's formal disaster-management framework.

13. Study links

- FACT: Foundation companion: `basic/26_Disaster-Management-Framework-and-Sendai.md`.
- FACT: `24_Coastal-and-Marine-Ecology-CRZ-Blue-Economy.md` - mangrove-based storm-surge buffering as ecosystem-based disaster-risk reduction.
- FACT: `07_Biosphere-Reserves-and-Ramsar-Sites.md` - wetlands' flood-buffering ecosystem service.
- FACT: `01_Ecosystem-Structure-and-Function.md` - the ecosystem-services framework underlying this topic's ecosystem-based risk-reduction integration.

CONSOLIDATED REGISTER NOTES

Disaster Management Framework and Sendai: HAZARD, EXPOSURE, VULNERABILITY, CAPACITY AND CYCLE MAP

1. **Hazard boundary:** A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk.
2. **Exposure boundary:** Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility.
3. **Vulnerability boundary:** Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure.
4. **Capacity boundary:** Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses.
5. **Risk relation:** Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone.
6. **Disaster boundary:** A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster.
7. **Prevention-mitigation distinction:** Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented.
8. **Preparedness boundary:** Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response.
9. **Response boundary:** Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster.
10. **Recovery boundary:** Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components.
11. **Build-Back-Better boundary:** Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability.
12. **Cycle-continuum boundary:** Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence.
13. **Domestic-law boundary:** India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework.
14. **NDMA-NEC boundary:** NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework.
15. **NIDM-NDRF boundary:** NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other.
16. **State-district boundary:** SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text.
17. **Sendai identity:** The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency.
18. **Priorities-targets boundary:** Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions.
19. **Global-domestic boundary:** Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction.
20. **Current evidence boundary:** Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence.

Disaster Management Framework and Sendai: PHASE, INSTITUTION, PRIORITY, TARGET AND JURISDICTION TRAPS

- Do not merge hazard with disaster.
- Do not merge exposure with vulnerability.
- Do not omit capacity from a risk analysis.
- Do not infer risk from hazard intensity alone.
- Do not call every hazard event a disaster.
- Do not merge prevention with mitigation.
- Do not call preparedness a response activity.
- Do not call emergency response long-term recovery.
- Do not reduce recovery to rebuilding identical assets.
- Do not make the disaster cycle a one-way sequence.
- Do not derive domestic legal powers from Sendai.
- Do not merge NDMA with NEC.
- Do not merge NIDM training with NDRF operations.
- Do not move national roles automatically to a DDMA.
- Do not call Sendai a binding treaty.
- Do not replace four priorities with seven targets.
- Do not replace seven global targets with four priorities.
- Do not equate global monitoring with domestic enforcement.
- Do not invent losses, indicators or institutional powers.
- Do not state a current plan or amendment status without dated official text.

Disaster Management Framework and Sendai: DISASTER-RISK-REDUCTION ANSWER SPINE

```
DEFINE HAZARD EXPOSURE VULNERABILITY CAPACITY RISK AND DISASTER
-> SEPARATE PREVENTION MITIGATION PREPAREDNESS RESPONSE AND RECOVERY
-> USE BUILD BACK BETTER TO REDUCE FUTURE RISK
-> MAP NDMA NEC NIDM NDRF SDMA AND DDMA ONLY BY OWNED ROLE
-> SEPARATE SENDAI FOUR PRIORITIES FROM SEVEN GLOBAL TARGETS
-> SEPARATE GLOBAL FRAMEWORK FROM DOMESTIC LAW INSTITUTIONS AND FUNDS
-> CONCLUDE WITH DATED LAW PLAN LOSS INDICATOR AND WARNING EVIDENCE
```

Disaster Management Framework and Sendai: LIVE LAW, PLAN, LOSS, INDICATOR, WARNING AND TARGET EVIDENCE BOUNDARY

UNDRR supplied substantive Sendai framework text. Indian official routes did not yield substantive current legal or plan text in this check, so no disaster loss, target progress, indicator, warning coverage, amended power or plan-status claim is asserted.

COMPLETE TOPIC ASCII MASTER FLOW DIAGRAM

High-yield use: Follow the panels as one topic-specific conceptual atlas: root question, classifications, processes or arguments, comparisons, objections and a final answer-writing spine. This text-native master is separate from both graphical flowchart artifacts.

ASCII MASTER FLOW - PANEL 1/12: Disaster-risk equation

```
HAZARD -> potentially damaging process or event
EXPOSURE -> people assets systems in harm's way
VULNERABILITY -> susceptibility to damage
CAPACITY -> resources and ability to manage risk
RISK -> interaction across a stated place and time
```

ASCII MASTER FLOW - PANEL 2/12: Hazard-to-disaster threshold

```
HAZARD OCCURS -> physical or human-induced event
EXPOSED SYSTEM -> contact with people assets or ecosystems
VULNERABILITY AND LOW CAPACITY -> losses amplify
COPING ABILITY SEVERELY TESTED -> serious disruption
DISASTER -> not every hazard crosses this threshold
```

ASCII MASTER FLOW - PANEL 3/12: Risk-reduction ladder

```
PREVENTION -> avoid creation or existence of risk where possible
MITIGATION -> reduce adverse impact of remaining risk
PREPAREDNESS -> ready plans warnings skills and resources
RESPONSE -> immediate life-saving and basic needs
RECOVERY -> restore and improve while reducing future risk
```

ASCII MASTER FLOW - PANEL 4/12: Build Back Better loop

```
DAMAGE AND NEEDS ASSESSMENT -> establish recovery baseline
REHABILITATION -> restore essential functions
RECONSTRUCTION -> rebuild assets and systems
RISK-INFORMED IMPROVEMENT -> avoid old vulnerability
FEEDBACK -> prevention mitigation and preparedness improve
```

ASCII MASTER FLOW - PANEL 5/12: Domestic architecture firewall

```
DISASTER MANAGEMENT ACT -> domestic legal base
NDMA -> national policy and guidelines
NEC -> executive coordination and implementation support
SDMA AND DDMA -> state and district statutory levels
RULE -> roles come from domestic law, not Sendai text
```

ASCII MASTER FLOW - PANEL 6/12: Training-response split

```
NIDM -> training research documentation capacity development
NDRF -> specialised disaster response operations
NDMA -> policy and guidance
DDMA -> district planning and implementation context
RULE -> institution type determines the answer
```

ASCII MASTER FLOW - PANEL 7/12: Sendai identity card

```
PERIOD -> 2015 to 2030
NATURE -> global voluntary non-binding framework
PURPOSE -> prevent new and reduce existing disaster risk
SCOPE -> people livelihoods health assets and systems
NOT -> Indian statute fund or response force
```

ASCII MASTER FLOW - PANEL 8/12: Four priorities rail

```
PRIORITY 1 -> understand disaster risk
PRIORITY 2 -> strengthen risk governance
PRIORITY 3 -> invest in resilience
PRIORITY 4 -> preparedness response and Build Back Better
RULE -> fields of action, not the seven global targets
```

ASCII MASTER FLOW - PANEL 9/12: Seven-target logic

```
TARGETS A TO D -> substantially reduce specified losses
MORTALITY AND AFFECTED PEOPLE -> human impact directions
ECONOMIC AND CRITICAL-SERVICE LOSS -> system impact directions
TARGETS E TO G -> substantially increase strategies cooperation warnings
RULE -> indicator values require official monitoring metadata
```

ASCII MASTER FLOW - PANEL 10/12: Global-domestic bridge

```
SENDAI -> global principles priorities and targets
INDIAN LAW -> authorities powers duties and funds
NATIONAL AND STATE PLANS -> domestic implementation choices
SECTOR AGENCIES -> warnings infrastructure health and relief
ALIGNMENT -> policy bridge, not legal identity
```

ASCII MASTER FLOW - PANEL 11/12: Evidence status screen

```
LOSS FIGURE -> event date definition and reporting source
TARGET PROGRESS -> indicator baseline and period
EARLY WARNING -> hazard geography reach and access
INSTITUTIONAL POWER -> current statutory provision
PLAN STATUS -> dated official edition or amendment
```

ASCII MASTER FLOW - PANEL 12/12: Disaster answer spine

```
DEFINE -> hazard exposure vulnerability capacity risk and disaster
SEQUENCE -> prevention mitigation preparedness response recovery
MAP -> NDMA NEC NIDM NDRF SDMA and DDMA by owned role
SEPARATE -> four Sendai priorities from seven global targets
VERIFY -> domestic law plan loss indicator and warning evidence
```